Pressure Normalization Produces an Entropy-One Prime-Orbit Law for a Hénon Instability Suspension

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Abstract

The raw instability roof on a certified area-preserving Hénon survivor has correct local Euler-factor amplitudes but an incorrect global convergence class. We give the canonical dynamical repair. Let h_* be the unique zero of the pressure $P(-s\tau)$ for the positive Hölder instability roof τ. The normalized roof $\widehat{\tau} = h_*\tau$ remains non-lattice and has suspension entropy exactly one. Since the symbolic base is mixing, the Parry–Pollicott prime orbit theorem applies and yields $\Pi(T) \sim e^T/T$. The normalized Euler atoms are exactly $\log P_\gamma\, P_\gamma^{-rs}$ for the intrinsic positive real labels $P_\gamma = |\Lambda_\gamma|^{h_*}$. Thus the construction matches the prime-number counting exponent and all repetition amplitudes without a fitted parameter. Its remaining obstruction is arithmetic rather than dynamical: no theorem makes these real labels rational primes, and no critical-line continuation or Hilbert–Pólya operator is obtained.

1 Inherited Hénon suspension

Let $H_6(q,p) = (1 - 6q^2 - p, q)$ and restrict it to the previously certified four-state hyperbolic survivor. The adjacency matrix A satisfies $A^4 > 0$, so its subshift is topologically mixing. The unstable Jacobian defines a positive Hölder roof

$$\tau(z) = \log J^u(z).$$

For a primitive orbit γ, its roof period is

$$\ell_\gamma = \sum_{z\in\gamma} \tau(z) = \log|\Lambda_\gamma|.$$

Prior exact algebra supplies two primitive multipliers whose logarithms are incommensurable. Hence τ is non-lattice. These facts are inherited with hashes frozen in the certificate.

For real s, let $P(-s\tau)$ denote topological pressure. Positivity makes it strictly decreasing, and the certified pressure theorem gives a unique root h_* satisfying

$$0.277980 < h_* < 0.277987. \tag{1}$$

2 Canonical pressure normalization

Theorem 2.1 (Entropy-one normalization)**.** *Define*

$$\widehat{\tau} = h_*\tau, \qquad \widehat{\ell}_\gamma = h_*\ell_\gamma.$$

Then $\widehat{\tau}$ is positive, Hölder, and non-lattice. Its suspension flow has topological entropy one.

Proof. Positive scalar multiplication preserves positivity and Hölder regularity. If the periods of $h_*\tau$ lay in a discrete subgroup, division by $h_* > 0$ would make τ lattice, contrary to the inherited theorem. The entropy h of a suspension roof is characterized by $P(-h\widehat{\tau}) = 0$. Here

$$P(-1 \cdot \widehat{\tau}) = P(-h_*\tau) = 0.$$

Uniqueness of the pressure root gives $h = 1$. $\square$

The normalization is canonical within the chosen roof: it is the unique positive scalar multiple whose suspension entropy is one. It is not chosen from a prime or zero table.

3 Prime orbit theorem

Let $\widehat{\Pi}(T)$ count primitive closed suspension orbits of least period at most T.

Theorem 3.1 (Dynamical prime-number law)**.** *For the pressure-normalized Hénon suspension,*

$$\boxed{\widehat{\Pi}(T) \sim \frac{e^T}{T}} \qquad (T \to \infty).$$

Proof. The base is a mixing finite-type shift, the roof is positive Hölder and non-lattice, and Theorem 2.1 gives entropy one. The prime orbit theorem for weak-mixing hyperbolic suspension flows therefore gives $e^{hT}/(hT)$ with $h = 1$. $\square$

This is the exact dynamical analogue of the prime number theorem at the counting level. Non-lattice is essential: a lattice suspension has periodic fluctuations rather than this continuous asymptotic.

4 Exact local labels and repetitions

Define the positive real orbit label

$$P_\gamma = e^{\widehat{\ell}_\gamma} = |\Lambda_\gamma|^{h_*}.$$

Then

$$1 - e^{-s\widehat{\ell}_\gamma} = 1 - P_\gamma^{-s}$$

and

$$\partial_s \log(1 - P_\gamma^{-s}) = \sum_{r \geq 1} \log P_\gamma \, P_\gamma^{-rs}. \tag{2}$$

Thus the pressure normalization simultaneously fixes the global counting exponent and preserves the exact local prime-power syntax identified in the preceding raw-clock audit.

Equation (2) is not an arithmetic prime correspondence. It uses one positive real label per orbit. Replacing P_γ by a nearby prime would be post-hoc fitting and is forbidden.

5 Analytic and arithmetic boundaries

The associated suspension zeta has its initial Euler-product domain $\Re s > 1$ and the standard entropy-one boundary singularity underlying Theorem 3.1. This removes the raw clock's absolute convergence on $\Re s = 1/2$. It does not prove meromorphic continuation to that line, a functional equation, gamma factors, or equality with the completed Riemann divisor.

The exact arithmetic question is now isolated:

$$\boxed{|\Lambda_\gamma|^{h_*} \stackrel{?}{\in} \mathbb{P}}$$

for a complete, multiplicity-preserving orbit family. No evidence in the pressure theorem addresses this statement. The first necessary audit is the algebraic type of the raw multiplier Λ_γ at every period.

6 Evaluator verdict

The outcome is a genuine all-period analytic structure, but not a rational- prime bridge. The strict Route-A tuple is

$$(A1_{\text{WEAK}}, A2_{\text{ANALYTIC}}, A3_{\text{PARTIAL}}, A4_{\text{FORMAL}}),$$

with overall `ROUTE_A_EXPLORATORY`. Route B is not authorized because no operator, self-adjoint domain, von Mangoldt trace identity, or xi determinant exists.

7 Conclusion

Pressure normalization is the first large repair in this batch: it converts the raw Hénon instability clock into an entropy-one non-lattice suspension with a true prime orbit theorem. The bridge is now blocked at a sharply arithmetic gate—whether its intrinsic real labels can ever be rational primes—rather than at orbit density or repetition bookkeeping.

References

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